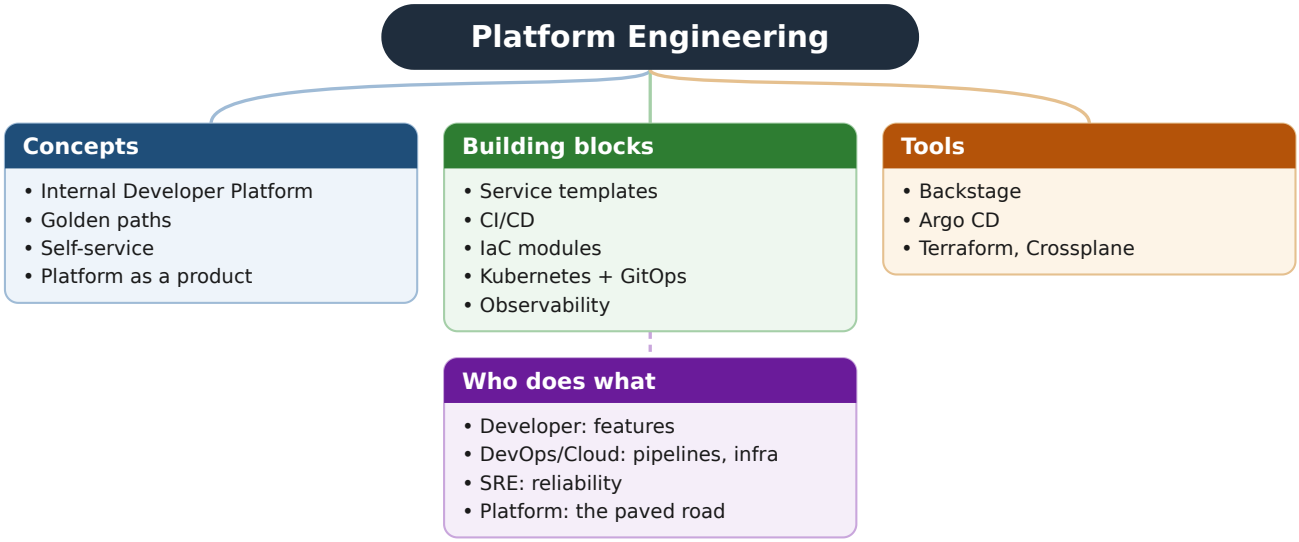
16. Platform Engineering

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16.1 Concepts

Term	Meaning
Platform engineering	Building internal tools and paved roads so developers can ship safely by themselves
Internal Developer Platform (IDP)	The self-service layer: templates, pipelines, environments, infrastructure, observability — all wired together
Developer experience (DevEx)	How easy and fast it is for developers to do their work
Self-service	Developers create what they need (new service, database, environment) without opening tickets
Golden path	The recommended, pre-built, supported way to do a common task (e.g. "create a new web service")
Platform as a product	The platform team treats developers as customers: docs, feedback, roadmap
Platform API	The interface developers use: a portal, a CLI, Git templates, or Kubernetes custom resources
Backstage	Open-source developer portal: software catalog, templates (scaffolder), docs (TechDocs)
Crossplane	Lets you create cloud infrastructure through Kubernetes-style APIs

16.2 DevOps vs SRE vs Platform Engineering

Role	Main focus
Developer	Builds application features
DevOps engineer	Automation, CI/CD, infrastructure, culture of shared ownership between dev and ops
Cloud engineer	Designs and runs cloud infrastructure (networks, IAM, compute, cost)
SRE	Reliability: SLOs, monitoring, incident response, reducing toil with engineering
Platform engineer	Builds the internal platform and golden paths that the other roles use

```
Developer ──uses──► Internal Developer Platform (portal + templates + pipelines)
                          │ built and run by Platform team
                          ├── CI/CD (GitHub Actions)        ← DevOps practices
                          ├── IaC modules (Terraform)       ← Cloud engineering
                          ├── Kubernetes + Argo CD (GitOps)
                          └── Monitoring, SLOs, on-call      ← SRE
```

16.3 What a golden path template contains

- Repo skeleton (app code, Dockerfile, tests, README).
- CI pipeline (build, test, scan, push).
- Helm chart / Kubernetes manifests with sensible defaults (probes, limits, non-root).
- Argo CD Application per environment.
- Monitoring dashboards and alerts.
- Security defaults (secrets via External Secrets, least-privilege IAM role).

16.4 Final project reference architecture

```
GitHub (app repo) ──► GitHub Actions: test → scan → build → push ──► Amazon ECR
                                          │ updates image tag
GitHub (config repo: Helm values) ◄──────┘
        │ watched by
Argo CD (in EKS) ──► deploys Helm chart ──► App pods (EKS, private subnets)
                                                  │
AWS (built with Terraform): VPC, subnets, NAT, EKS, ECR, IAM roles, ALB
Observability: Prometheus + Grafana (+ CloudWatch) — alerts on SLOs
```

REMEMBER

- A good platform makes **the right way the easy way**.